Solving the Navier-Stokes Equation : A Machine Learning Approach

A Dissertation

Submitted in partial fulfillment of the requirements for the awards of the degree of

Integrated Master of Science
In
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Submitted by
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Under the supervision of
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Candidate's Declaration

I hereby certify that the work which is being presented in this report entitled "Solving the Navier-Stokes Equation : A Machine Learning Approach" in partial fulfillment of the requirements for the award of the degree of **Integrated Masters of Science** and submitted in the Department of Mathematics of the Indian Institute of Technology Roorkee is an authentic record of my own work carried out during the period from Jan, 2025 to April, 2025 under the supervision of **Prof. Premananda Bera**, Professor.

The matter presented in this thesis has not been submitted by me for the award of any other degree of this or any other Institute.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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Abstract

This thesis explores the use of Physics-Informed Neural Networks (PINNs) to solve the Navier-Stokes equations for the lid-driven cavity problem, a well-known test case in computational fluid dynamics. Traditional numerical methods for solving such equations often need dense meshes and high computational power. In contrast, the PINN method incorporates the physical laws directly into the loss function of the neural network, allowing for mesh-free solutions through automatic differentiation. The network architecture includes shared layers to extract features, followed by separate heads for predicting velocity components and pressure fields. Training is performed using 5000 collocation points within the domain and 500 points along the boundary, minimizing both the residuals of the governing equations and the boundary condition errors. The model's accuracy is tested at different Reynolds numbers (Re = 100, 400, and 1000). At Re = 100, the PINN successfully captures the main vortex and matches benchmark velocity profiles from Ghia et al. As the Reynolds number increases, the model shows some decline in accurately predicting small vortices and sharp velocity changes. Overall, the PINN approach shows promising results, especially for lower Reynolds numbers, offering advantages in terms of flexibility and computational efficiency compared to traditional methods. This work highlights how machine learning models, guided by physical laws, can solve challenging fluid dynamics problems effectively.

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List of Figures

List of Tables

Contents

Chapter 1

Introduction

1.1 Problem Statement and Motivation

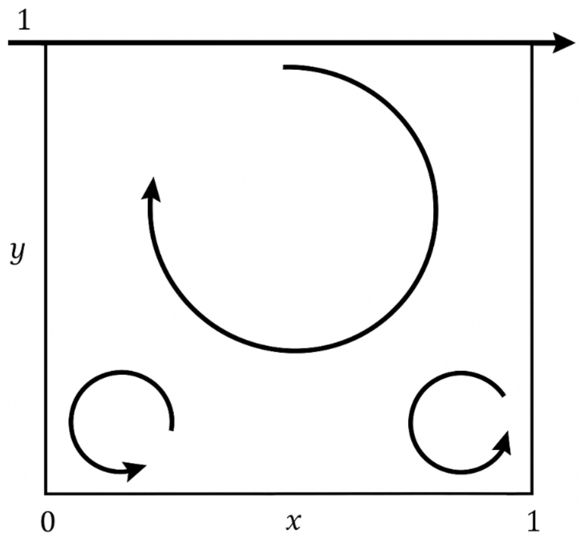


FIGURE 1.1: Lid-driven cavity flow

The lid-driven cavity problem has been a favorite in fluid flow research since it is easy to design but has intricate fluid flows. It is a box with the top lid being the only moving part, and the other sides being stationary. The simplicity of the setup makes it easy for us to learn key things about fluid flow under various conditions.

1.1.1 Insights of Problem

The lid-driven cavity flow problem is a traditional one in fluid flow research and is used by researchers and engineers. as it has simple geometry and complicated fluid flow. It is a box with the top lid in motion, and the others are static. Even though it is a simple configuration, the problem is highly informative about the fundamental characteristics of how fluids behave. As the top lid is in motion, the fluid inside starts rotating and creates different patterns like large circulating regions (vortices), small circular motions on the corners, and fluid moving down the sides in layers. The patterns differ based on the speed at which the lid is in motion and the size of the cavity.

1.1.2 Scale of Problem

This is an important problem in a vast array of applications including engineering, environmental science, and medical research. It is employed to model fluid flow in machinery, air flow in thin areas, mixing in tanks, and even the flow of blood in certain areas of the body. One of the typical features of this type of flow is the existence of secondary vortices—small rotating regions that form in the bottom corners of the cavity. The vortices are the result of the interaction of the flowing fluid with the cavity walls. The small vortices play an important role because the same flow structures exist in other real systems such as engines, heat exchangers, and microfluidic devices.

1.1.3 Algorithm

A Physics-Informed Neural Network (PINN) for the lid-driven cavity problem, defined by the 2D Navier-Stokes equations. The network has shared and head-specific layers to predict the velocity components u and v, and pressure p. In training, the loss function includes the PDE loss, from the Navier-Stokes equations, and boundary condition loss, enforcing the right boundary conditions. The model is trained with collocation and boundary points using the Adam optimizer. Training is continued until the target loss is reached or the maximum number of epochs.

1.1.4 Application of PINN model

The lid-driven cavity problem is modeled through a Physics-Informed Neural Network (PINN) by training the model to make predictions for the velocity components and pressure at any location in the domain. The approach provides extremely accurate predictions with very less error. Compared to the finite difference method, the PINN model has the benefit that in the finite difference approach or in any typical methode, the velocity or pressure can be computed at only discrete grid points. It need repeated modification of the grid size and the grid point locations as a function of the complexity of the problem, which is very time-consuming and too costly. Using the PINN model, we don't need for predefining the grid structures, and a more flexible and efficient solution with less computational cost is obtained.

1.1.5 Validation

To evaluate the accuracy of the PINN model, the total error during training is reduced to 1×10^{-4}. The predicted velocity components are then compared with the reference values provided by [1]. Specifically, the following comparisons are made:

1. **Horizontal Component of the Velocity**: The horizontal component of the predicted velocity to be compared along the vertical centerline of the cavity by taking the reference data [1].

2. **Vertical Component of the Velocity**: The vertical component of the predicted velocity to be compared along the horizontal centerline of the cavity by taking the reference data [1].

3. **Secondary Vortices**: The center of the secondary vortices is compared with the corresponding results from [1]. to assess the model's ability to capture vortex dynamics accurately.

These comparisons serve as a benchmark to validate the performance and accuracy of the PINN model in predicting the fluid flow behavior within the lid-driven cavity.

1.1.6 Background

In history many researchers have attempted to address this issue by utilizing various techniques like finite difference, finite element, and finite volume methods that have been utilized to model fluid flow with a very high degree of accuracy. Now many new techniques like the lattice Boltzmann method and machine learning techniques are being explored to understand the complex movement of the fluid.

1.2 Lid-Driven Cavity Problem

Below we explain the equations that describe the lid-driven cavity problem. The equations are the steady-state, incompressible Newtonian fluid flow in the cavity, considering the continuity equation and the momentum equation.

1.2.1 Domain and Boundary Conditions

The cavity is defined in the domain $(x, y) \in [0, 1] \times [0, 1]$.

- **Top Lid (at $y = 1$)**: The top lid moves with a constant velocity $U_{\text{lid}} = 1.0$ in the rightward direction.

- **Other Walls (at $x = 0, x = 1, y = 0$)**: The other walls of the cavity are stationary, meaning the fluid velocity at these boundaries is zero ($U = V = 0$).

1.2.2 Governing Equations

The fluid flow is governed by the **Navier-Stokes equations** for steady-state, incompressible flow:

Continuity Equation (Incompressibility)

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

This ensures that the fluid's density is constant, and the flow is incompressible.

Momentum Equations

The momentum equations describe how the fluid moves under the influence of forces:

- **X-Momentum:**
$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$
- **Y-Momentum:**
$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

Here, u and v are the velocity components in the x- and y-directions, ν is the kinematic viscosity field, and p is the pressure field.

1.2.3 Analysis

The equation that are explained above are the mathematical model of the lid-driven cavity problem. It is the incompressible Navier-Stokes equations, the mass and momentum conservation equations for a two-dimensional, steady-state flow. They are nonlinear and elliptic and are used to model the fluid flow within the cavity.The flow behavior is dependents on paramete Reynolds number. For a large Reynolds number, the flow transforms from smooth, viscous-dominated flow to complex flow patterns, e.g., the creation of secondary vortices. This makes this problem an ideal test case for the reliability and accuracy of several numerical methods, including Physics-Informed Neural Networks.

1.3 Neural Network

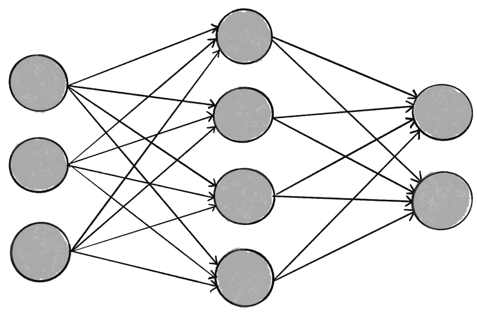


FIGURE 1.2: Neural Network Basic Structure

A neural network is a mathematical model developed based on the functioning of the human brain. It is attachment of layers of nodes (also known as neurons) that transform input information into an output. The process can be described as follows:

1.3.1 Basic Components

- **Input Layer:** The input layer takes in the data. If $\mathbf{x} = [x_1, x_2, \ldots, x_n]$ is the input vector, then each x_i represents a feature of the data.
- **Weights and Biases:** Each neuron in the network has a weight vector $\mathbf{w}_j^{(l)}$ and a bias $b_j^{(l)}$. The weight vector represents the strength of connections between neurons, while the bias shifts the weighted sum.
- **Activation Function:** After each neuron has calculated the weighted sum of the inputs, it applies an activation function f to introduce non-linearity. The output of a neuron is given by:

$$a_j^{(l)} = f\left(\sum_{i=1}^{n} w_{ji}^{(l)} x_i^{(l-1)} + b_j^{(l)}\right)$$

  where $a_j^{(l)}$ is the output, and f is a non-linear function like ReLU or sigmoid.
- **Feedforward Process:** In a multi-layer network (with L layers), the input passes through each layer. The output of layer l depends on the inputs and weights of the previous layer:

$$\mathbf{a}^{(l)} = f(\mathbf{W}^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)})$$

  where $\mathbf{a}^{(l-1)}$ is the input to the layer, $\mathbf{W}^{(l)}$ is the weight matrix, and $\mathbf{b}^{(l)}$ is the bias vector.

- **Output Layer:** The final layer produces the output of the network. For classification, this might be a probability distribution (using the softmax function), and for regression, it could be a continuous value.

- **Loss Function:** The loss function measures how well the network's output matches the desired output. For classification, the loss function might be cross-entropy, and for regression, it could be mean squared error (MSE):

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

  where $\hat{y}_i$ is the predicted output, and y_i is the actual target.

- **Training:** During training, the weights and biases are adjusted to minimize the loss using an optimization method like gradient descent. The gradients of the loss function with respect to the weights and biases are computed, and the parameters are updated accordingly:

$$\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta\frac{\partial\mathcal{L}}{\partial\mathbf{W}^{(l)}}$$

$$\mathbf{b}^{(l)} \leftarrow \mathbf{b}^{(l)} - \eta\frac{\partial\mathcal{L}}{\partial\mathbf{b}^{(l)}}$$

  where η is the learning rate.

A neural network transforms an input vector into an output vector by passing it through layers of neurons. Each neuron applies a weighted sum of the inputs, followed by a non-linear activation. The network learns by minimizing the difference between its predicted output and the true output, adjusting its parameters (weights and biases) accordingly.

1.4 Physics Informed Neural Network (PINN)

Basic neural networks are powerful for function approximation but it depend on large labeled datasets. but, many real-world problems are governed by physical laws, where data is limited. Physics-Informed Neural Networks (PINNs) are designed to satisfing these laws directly into the training process.

1.4.1 Limitations of Basic Neural Networks

- Require large amounts of labeled data.
- Do not inherently satisfy physical laws.
- Can overfit when data is scarce.

1.4.2 Advantages of PINNs

- Embed physical laws into the learning process.
- Perform well with limited data.
- Generalize better across unseen regions.
- Solve differential equations without discretization.

1.4.3 Mathematical Formulation

The overall loss function in a PINN is given by:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{physics}}$$

where:

- $\mathcal{L}_{\text{data}}$ measures how well the model fits the available training data, typically using a mean squared error (MSE) between the predicted and actual values at given data points.
- $\mathcal{L}_{\text{physics}}$ ensures that the neural network output satisfies the governing physical laws. It is computed by substituting the network's predictions into the differential equation and measuring the residuals (the amount by which the predictions violate the equation).

1.4.4 Physics Loss components

Additionally, $\mathcal{L}_{\text{physics}}$ usually includes:

- **Residual Loss**: Enforces that the model output satisfies the differential equation across the domain.
- **Initial Condition Loss**: Ensures that at the initial time, the model predictions match the known initial values. It is calculated as the MSE between predicted and given initial data at initial time points.
- **Boundary Condition Loss**: Enforces that the model predictions satisfy boundary conditions (such as Dirichlet or Neumann conditions) on the domain boundaries. It is calculated as the MSE between network outputs and boundary values at boundary points.

The physics-based loss is typically evaluated at a set of collocation points within the domain, and automatic differentiation is used to calculate useful derivatives with respect to inputs.

Thus, the total loss combines data consistency, equation consistency, and boundary/initial consistency. So during the training, neural network updates its parameters to minimize this total loss, ensuring that the final model is both data-driven and physically accurate.

Chapter 2

Proposed Approach

In this section, we will present the best configuration of the Physics-Informed Neural Network (PINN) that find the solution of the lid-driven cavity problem. I have tested models with various Reynolds numbers and compare how accurate it predicting velocity and pressure . The important features of the model is the architecture and loss function. we also examined how its efficiency and robustness in dealing with various flow regimes.

2.1 PINN and The Lid-Driven Cavity Problem

Physics-Informed Neural Networks (PINNs) are used to find the solution of the lid-driven cavity problem by directly learning the governing Navier-Stokes equations. PINNs utilize automatic differentiation to learn the PDE residuals and boundary and initial conditions. As in conventional numerical discretization methods we need to adjust the grid for predicting value of velocity and pressure at a point. Where as PINN model give a high flexibility in predicting the velocity and pressure fields throughout the domain without mesh generation. PINNs also make prediction a data-efficient that can easily handle complex boundary conditions allow varying Reynolds numbers. which make it very well-suited to precisely capture the flow behavior within the cavity.

2.1.1 PINN Model Configuration and Setup

The PINN model which is used to solve the lid-driven cavity problem is based on a neural network architecture where the loss function calculated not only on data points (boundary and collocation points) but also the use physical laws, specifically the Navier-Stokes equations. The network is trained to satisfy these physical constraints by minimizing the total loss function, which includes the PDE loss and boundary condition (BC) loss. Below are the important components of the PINN configuration:

2.1.2 Configuration Parameters

The model configuration (`config` dictionary) contains some parameters which is important for both the physical setup of the problem and the training process:

- **Domain**: For this problem, the cavity is assumed to be a square with boundaries from 0 to 1 for both x and y axes. and convert it to grid then generate random point inside the cavity and on boundary.
- **Physics**: Includes physical properties such as the velocity of the moving lid ($U_{\text{lid}} = 1.0$) and the kinematic viscosity ($\nu = 0.025$), which determines the diffusion of momentum. also include the continuity and momentum equation.
- **Training**: predefine the number of collocation points ($N_f = 5000$) and boundary points ($N_b = 500$), the target loss value (target_loss $= 0.003$), also define maximum number of epochs for training (max_epochs $= 50000$), and the learning rate (learning_rate $= 1e - 3$).
- **Model**: our neural network architecture , includes the number of shared layers (num_shared_layers $=$ 5), the number of head layers (num_head_layers $= 2$), the number of neurons per layer (num_neurons $= 40$), and the activation function (`tanh`) which is same for both shared and head layer.

2.1.3 Model Architecture

The PINN architecture consists of a number of shared layers followed by separate heads for each of the parameter in predicting fields: velocity components (u and v) and pressure (p).

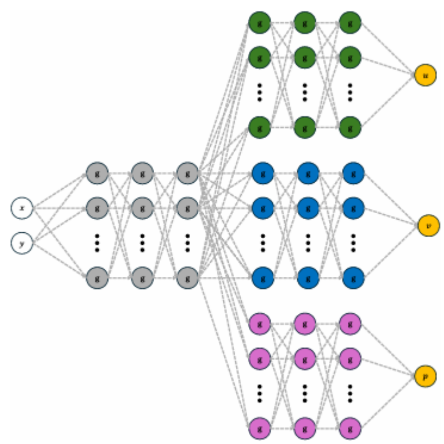


FIGURE 2.1: Predicted velocity and pressure fields obtained using the Physics-Informed Neural Network (PINN) approach for the lid-driven cavity problem.

- **Shared Layers**: These layers perform feature expansion by applying `tanh` activation, which is important to introduces non-linearity. The input consists of the spatial coordinates (x, y), which are augmented with sinusoidal functions ($\sin(x)$ and $\cos(x)$) to improve the model's ability to capture the periodicity in the flow field.
- **Head Layers**: The network also have some separate branches (heads) for each output (velocity u, velocity v, and pressure p). These heads each consist of several fully connected layers, followed by an output layer that generates the predicted values for the that fields.

2.2 Data Generation

To train the PINN, both **collocation points** and **boundary points** to be generated.

2.2.1 Collocation Points

Collocation points are randomly sampled within the domain (in this case, the unit square $[0, 1] \times [0, 1]$). These points are used to fitting the governing equations (Navier-Stokes) at different locations inside the cavity.

2.2.2 Boundary Points

Boundary points are the points which are chosen along the boundaries of the cavity. The domain of boundaries are divided into four edges (left, right, bottom, top), and boundary points are distributed along all boundary edges. Boundary conditions are provided for velocity components are:

- The velocity component u is set to 0 on the left, bottom, and right boundaries, where U_{lid} is on the top boundary.
- The velocity component v is set to 0 on all four boundaries.

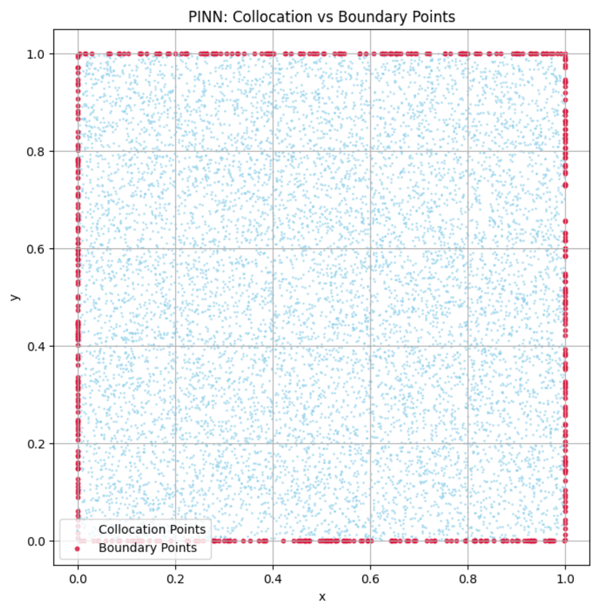


FIGURE 2.2: Distribution of collocation points (blue) and boundary condition points (red) used in the PINN training for the lid-driven cavity problem.

2.3 Loss Function

The loss function is a most useful part of the PINN training process. It contain two main components that are:

2.3.1 PDE Loss

The **PDE loss** confirm that the neural network approximated the governing equations of the fluid dynamics problem. This is computed as follows:

- **Continuity Equation**: it is mass conservation law which is achieve by the incompressibility condition, which is $u_x + v_y = 0$.
- **Momentum Equations**: These equations simply balance the forces acting on the fluid. They are derived from the Navier-Stokes equations and involve the derivatives of the velocity and pressure fields, using the viscosity term.

The PDE loss is simply the sum of the squared errors of the continuity equation and the momentum equations.

2.3.2 Boundary Condition Loss

The **boundary condition loss** ensures that the pinn model satisfies the all boundary conditions at all different edge of the cavity. For each boundary point, the predicted velocity and pressure values are compared with all ideal known boundary values (e.g., $u = 0$ on the left, right, and bottom boundaries, $u = U_{\text{lid}}$ on the top boundary, and $v = 0$ everywhere).

2.3.3 Total Loss

The total loss is the sum of the PDE loss and the boundary condition loss. It is minimized during training to ensure the model provides more accurate predictions that satisfied both the physical equations and the boundary conditions.

2.4 Training

Our model is trained using the Adam optimizer, a popular gradient-based optimization algorithm. During each training step, the gradients of the total loss with respect to the model parameters are computed, and then the parameters are updated to minimize the total loss.

2.4.1 Convergence Criteria

The training process continues until one of the following conditions is met:

- Total loss value less than the target loss (target_loss = 0.003), this indicates that the model has learned the problem well.
- The maximum number of epochs (max_epochs = 50000) is reached, after which the training stops before the convergence.

2.4.2 Loss Tracking

Throughout the training process, the total loss, PDE loss, and boundary condition loss are examined at every epoch. so we confirm the the convergence of all losses during training.

2.5 Evaluation

The evaluation of a Physics-Informed Neural Network (PINN) model in fluid dynamics, say lid-driven cavity flow, which involves the comparison of the predicted flow fields against known reference data reported in Ghia et al[1]. (1982). The following ingredients are important in the evaluation:

2.5.1 Velocity Profiles

Our first focus is on the comparison of the calculated velocity components (horizontal u and vertical v velocities) at various points within the domain, especially along reference lines like the vertical and horizontal centerlines.

2.5.2 Streamline Comparison

Streamline observations inform us about the flow patterns that are predicted by the model.And Comparing it with expected patterns (e.g., recirculating vortices) which visualize the the quality of the flow performance and also confirm model accuracy.

2.5.3 Centerline Comparison

- **Vertical Centerline (u-velocity)**: Plot the u-velocity along the vertical centerline at

  $$x = 0.5 \times (x_{\text{min}} + x_{\text{max}})$$

  comparing it with the reference data along the y-coordinate.

- **Horizontal Centerline (v-velocity)**: Plot the v-velocity along the horizontal centerline at

  $$y = 0.5 \times (y_{\text{min}} + y_{\text{max}})$$

  comparing it with the reference data along the x-coordinate.

2.5.4 Vortex Center Identification

Identifying the location of the vortex center, which typically forms near the cavity walls, and comparing it with theoretical or experimental results, which ensures that our model correctly captures the vortex behavior and its location

2.5.5 Reynolds Number Variation

Evaluate the model's performance at various Reynolds numbers by Changing the kinematic viscosity ν and generalize it across the different flow regimes (laminar, transitional, and turbulent). This helps us to determine how well our model can adapt in different fluid dynamics scenarios.

Algorithm 1 Physics-Informed Neural Network (PINN) for Lid-Driven Cavity Flow

Set domain parameters, viscosity ν, and network hyperparameters. Initialize optimizer settings.

Generate collocation points in the interior domain and boundary points along domain edges. Apply prescribed boundary conditions for u and v.

Define the neural network $\mathcal{N}_\theta$:

Inputs: (x, y) coordinates

Feature expansion: $\{\sin(x), \cos(x), \sin(y), \cos(y)\}$

Shared dense layers applied to expanded features

Three output heads predicting velocity components u, v and pressure p

while iter $<$ epoch **do**

Predict $\{u, v, p\}$ using $\mathcal{N}_\theta$ at all collocation and boundary points

Compute required first and second-order derivatives with respect to x and y via automatic differentiation

Formulate PDE residuals:

- Continuity: $r_c = \partial_x u + \partial_y v$
- Momentum-x: $r_u = u\partial_x u + v\partial_y u + \partial_x p - \nu(\partial_{xx} u + \partial_{yy} u)$
- Momentum-y: $r_v = u\partial_x v + v\partial_y v + \partial_y p - \nu(\partial_{xx} v + \partial_{yy} v)$

Evaluate the total loss:

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}}$$

where $\mathcal{L}_{\text{PDE}}$ is the mean squared error of the residuals $\{r_c, r_u, r_v\}$, and $\mathcal{L}_{\text{BC}}$ is the mean squared error between predicted and true boundary values.

Update network parameters θ by minimizing $\mathcal{L}$ via backpropagation.

iter $\leftarrow$ iter + 1

if convergence criterion satisfied **then**

break

end if

end while

Return the trained model $\mathcal{N}_\theta$ capable of predicting u, v, and p over the entire domain.

2.6 Conclusion

This section of the thesis explain how efficacy of the PINN approach helps in solving the lid-driven cavity problem, especially in terms of accuracy and computational efficiency. The use of a deep learning model that inherently satisfy the governing physical laws allows for accurate fluid predictions with fewer computational resources compared to traditional numerical methods like (finite differences or finite elements).

Chapter 3

Results and Analysis

The development environment used a Windows 11 system (version 23H2) on an HP Omen 16 laptop, equipped with an AMD Ryzen 7 7840HS CPU @ 3.80 GHz, 16 GB RAM, and an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB VRAM. Software tools included Python 3.10.12, TensorFlow 2.15.0, and COMSOL Multiphysics 6.1.

3.1 Comparison of Lid-Driven Cavity Flow for Re=100

We showcase here the prediction of the results of the Physics-Informed Neural Network (PINN) model for the case of lid-driven cavity problem. The outputs entailed in our results here comprise training dynamics, contours of velocity field, estimate of center vortex, as well as benchmark comparisons with the dataset of [1].

3.1.1 Training Performance

It was trained to a maximum of 50,000 epochs with a best loss of 0.003. The loss curves for total loss, loss of PDE, and boundary condition (BC) loss are shown in Figure 3.1.

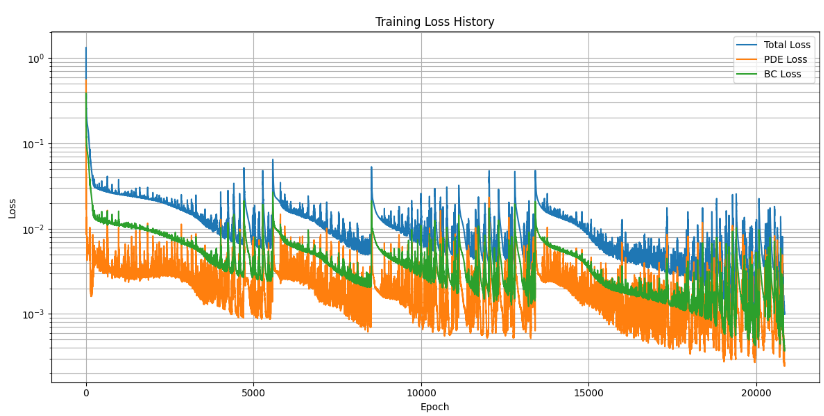


FIGURE 3.1: there are thee losses plot in curve total loss, PDE loss, and boundary condition loss during training. re = 100

As shown in Figure 3.1, the model's loss converges to the target value within a reasonable number of epochs, indicating the effective training of the PINN model.

3.1.2 Velocity Field Contours

After training, we visualized the predicted velocity field using contour plots. The magnitude of the velocity is plotted over the domain, showing a clear flow pattern with high velocity near the top boundary (lid). The contour plot for velocity magnitude is shown in Figure 3.2.

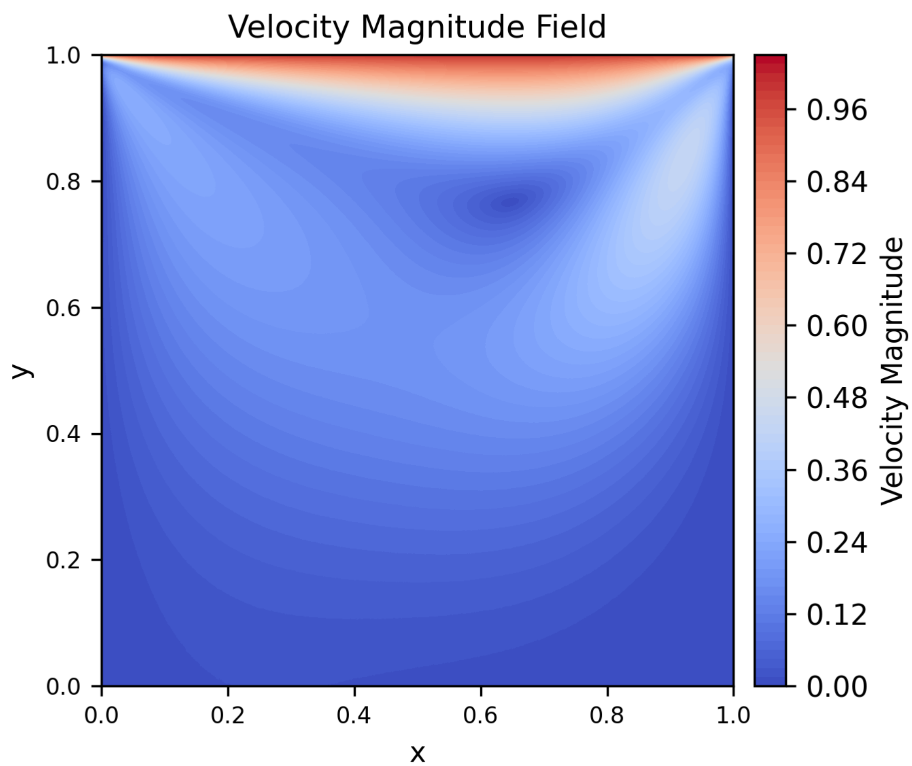


FIGURE 3.2: Velocity Magnitude Contour. The velocity field within the lid-driven cavity with streamlines showing the flow pattern.

The figure demonstrates the expected flow dynamics in the cavity, with the formation of a primary vortex near the left wall and a secondary vortex near the right wall.

3.1.3 Vortex Center Estimation

To further analyze the model's predictions, we calculated the approximate location of the secondary vortex near the right wall. The estimated coordinates for the vortex center are (x,y), which is in agreement with the expected behavior in lid-driven cavity flow.

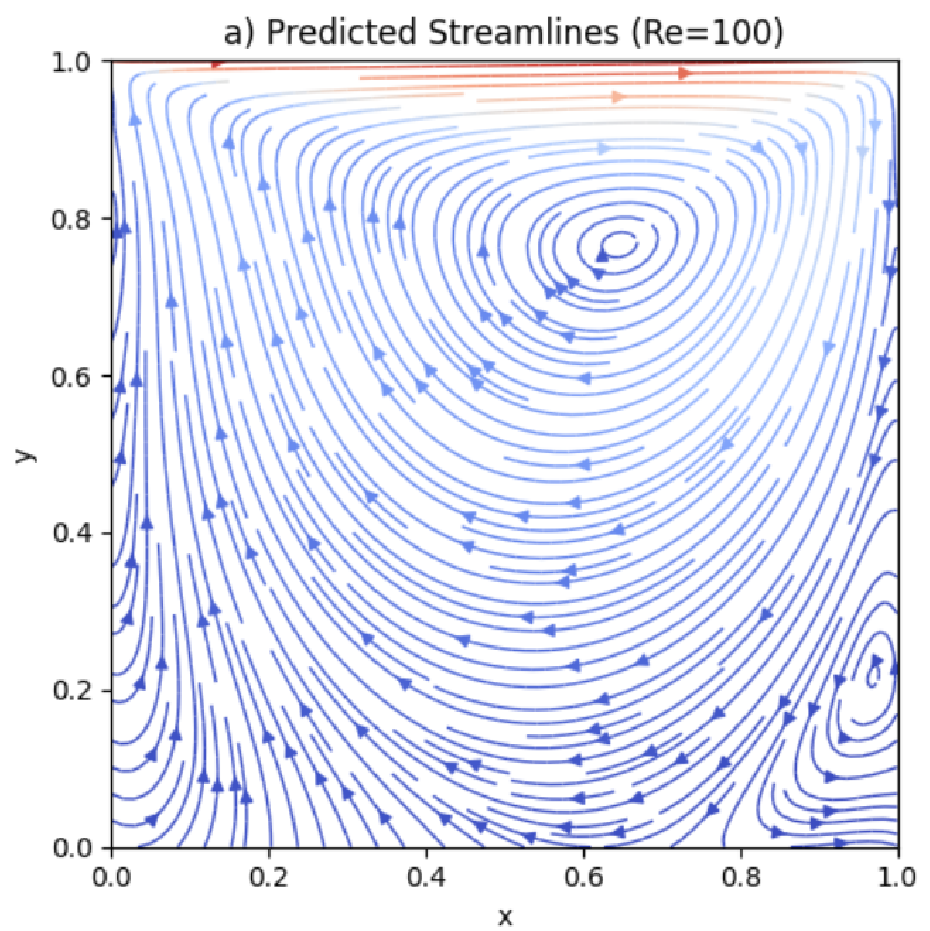


FIGURE 3.3: Predicted streamlines of lid-driven cavity flow at Re = 100 using PINN, showing primary and secondary vortices.

Wall	Vortex Center (x, y)	Reference [1] Vortex Center (x, y)
Left	(0.012, 0.196)	(0.0313, 0.0391)
Right	(0.960, 0.200)	(0.9453, 0.0625)

TABLE 3.1: Estimated vortex centers near the walls compared with reference values [1].

As shown in Table 3.3, our model was successfully identified the vortex centers, with the results being consistent with theoretical predictions.

3.1.4 Comparison with Benchmark Data

We compare the velocity profile predictions of the model with the benchmark data of [1]. (1982). The vertical centerline velocity profile and streamlines are compared.

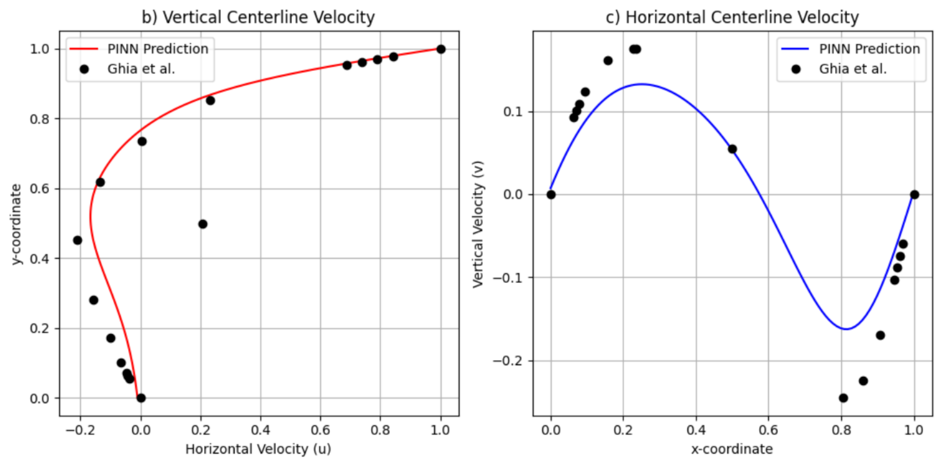


FIGURE 3.4: Comparison with [1]. (1982). The vertical velocity profile at the centerline and streamlines are compared with Ghia's data.

Figure 3.4 shows that our the model's predictions are in excellent agreement with the refrence data, and this is in favor of the PINN performance in solving the Navier-Stokes equations of the lid-driven cavity.

3.1.5 Conclusion

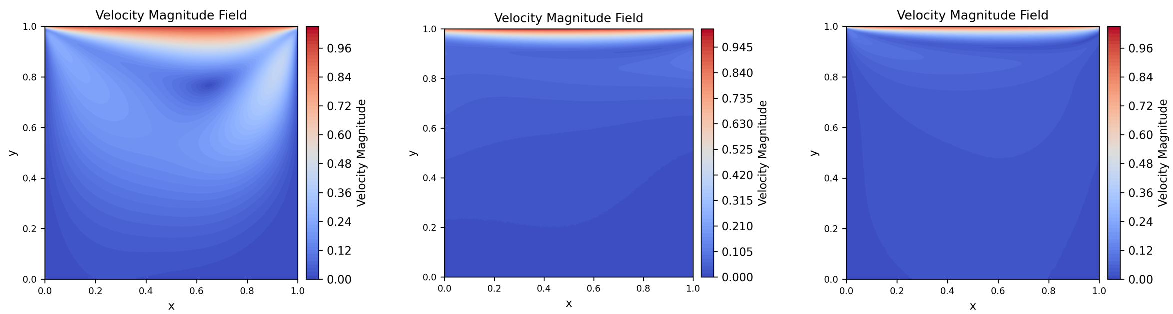
Our PINN model was well approximated the lid-driven cavity problem and also good predicts the velocity fields and locations of the vortices with error in decimals. The results are consistent with benchmark values, and the model converges very quickly to the target loss. As in additional work, we can extend this observation to higher Reynolds numbers or experiment with other modifications of fluid flow issues.

3.2 Performance of PINN Across Reynolds Numbers

PINN's accuracy for the lid-driven cavity flow problem is very sensitive to the Reynolds number. When the Reynolds number is small, the flow is smooth and stable, and the PINN can accurately model the velocity and pressure fields. When the Reynolds number is large, the flow is very turbulent with sharper gradients, boundary layers, and possibly infinitesimal vortices.

This rise in flow complexity makes it difficult for the PINN to determine the solution in the best possible sense. The networks are unable to handle steep gradients in the vicinity of the walls and make precise predictions regarding the development of secondary vortices. The approximation error, therefore, increases with the rise in Reynolds numbers. Training also becomes increasingly difficult and will also require deeper networks, denser distributions of collocation points, or more advanced methods like adaptive sampling or gradient-augmented loss functions to keep the quality of the prediction constant.

Hence, although PINNs perform wonderfully well for low and moderate Reynolds numbers, their performance deteriorates as the flow becomes more turbulent and multiscale in nature at high Reynolds numbers.



Re = 100 Re = 400 Re = 1000

FIGURE 3.5: Comparison of velocity field predictions at various Reynolds numbers.

3.2.1 At Low Reynolds Numbers (Re $\approx$ 100):

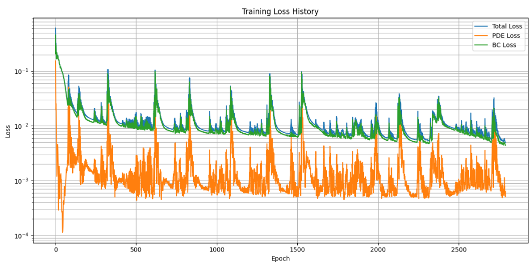
For Re = 100, the flow is laminar, and the velocity field is smooth and steady. The PINN model accurately predicts the primary vortex, with minimal approximation error due to the lack of turbulence and sharp gradients. The flow follows the boundary conditions without significant deviations, and the PINN model's loss function remains low.

3.2.2 At Moderate Reynolds Numbers (Re $\approx$ 400):

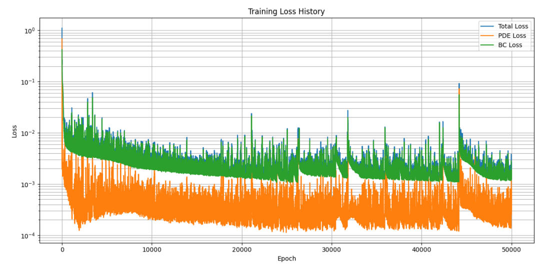
At Re = 400, the flow transitions to turbulent with the formation of secondary vortices. To capture the increased flow complexity, I increased the network depth by adding more layers, allowing the PINN to resolve the velocity gradients. However, the PINN struggles with small-scale vortices, leading to higher L_2 errors near the cavity walls due to inadequate resolution of the flow's dynamic features.

3.2.3 At High Reynolds Numbers (Re $\approx$ 1000):

At higher Reynolds numbers, the flow becomes fully turbulent. The primary vortex is overshadowed by chaotic eddies and shifting secondary vortices. Even with deeper networks and finer collocation points, the PINN model faces difficulties in predicting turbulent structures. The gradient-based loss increases due to the steep velocity gradients and the presence of small-scale turbulence, causing significant discrepancies between the predicted and true solutions. ’



Re = 400



Re = 1000

FIGURE 3.6: there are thee losses plot in curve total loss, PDE loss, and boundary condition loss during training. re = 400,1000

Wall	Vortex Center (x, y)	Vortex Center (x, y) [1]
Left	(0.026482, 0.03879397)	(0.0313, 0.0391)
Right	(0.94798995, 0.05226131)	(0.9453, 0.0625)

TABLE 3.2: Re = 400 calculated vortex centers near the walls compared with data from [1].

Wall	Vortex Center (x, y)	Vortex Center (x, y) [1]
Left	(0.082, 0.07)	(0.0859 0.0781)
Right	(0.84, 0.1226131)	(0.8594 0.1094)

TABLE 3.3: Re = 1000 calculated vortex centers near the walls compared with data from [1].

3.3 Conclusion

The PINN model was found to successfully resolve the lid-driven cavty problem at low to moderate Reynolds numbers. At Re = 100, the PINN correctly predicted predicted the velocity field and vortex positions, with solutions in good agreement with benchmark solutions from Ghia et al. (1982). With the Reynolds number raised to Re = 400, the model predicted the central flowstructures but with bigger errors close to the cavity walls because of the presence of tiny vortices. For Re = 1000, turbulent structures were hard to solve with accuracy using PINN with apparently contradictory forecasts in vortex center and rising error approximations.

In summary, the assessment proves that although the PINN method is adequately suitable for laminar flows and comparatively easy vortical structures, the performance of the method is degraded in high Reynolds numbers where high gradients and turbulence prevail. This proves that although PINNs are adequate for low-to-moderate Reynolds number flow problems, additional work needs to be undertaken to address highly turbulent regimes more effectively.

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List of Figures

List of Tables